

Davide Belli

STAFF ML RESEARCH ENGINEER

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Work Experience

Qualcomm AI Research

Amsterdam, NL

STAFF RESEARCH ENGINEER

Nov. 2019 - Present

- Designed and implemented novel Deep Learning methods for **Perception** (visual and RF domains), **Model Efficiency**, and **Agentic AI** (LLMs):
 - Invented a personalized biometric NN classifier improving accuracy from 90% to 99% AUC on challenging real-world data.
 - Implemented Neural KF and WLS methods for GNSS localisation, reducing positioning error by 50% in noisy urban canyons.
 - Designed a dynamic pruning algorithm for recent LLMs (Phi 3, Llama 3, ...), improving throughput by 40% and memory footprint by 45%.
 - Introduced structured representations and architectures in AI Agents, reducing model footprint by 30x at the same task accuracy.
- Integrated solutions in **products released to customers** (Samsung, Google) and **internal departments**. Filed 11 patent applications.
- Skills and technologies: Python (PyTorch, Hugging Face, vLLM, NumPy, Matplotlib, Pandas, Hydra), Bash, Git, Docker, Run:ai.

University of Amsterdam

Amsterdam, NL

GRADUATE TEACHING ASSISTANT

Oct. 2018 - Jun. 2019

- Graduate Teaching Assistant for the core courses Machine Learning 1, Deep Learning and Information Retrieval in the MSc AI at UvA.
- Held lab sessions, prepared and corrected exams, homework and lab assignments.

Education

University of Amsterdam

Amsterdam, NL

MSc IN ARTIFICIAL INTELLIGENCE

Sep. 2017 - Sep. 2019

- Graduated **Cum Laude**, GPA 8.8/10 (A+)
- Main Courses: Machine Learning, Computer Vision, NLP, Deep Learning, Reinforcement Learning, Information Retrieval
- Skills and technologies: Python (NumPy, Matplotlib, PyTorch, scikit-learn), MATLAB, Git, LaTeX

University of Trento

Trento, IT

BSc IN COMPUTER SCIENCE

Sep. 2014 - Jul. 2017

- Graduated **Cum Laude**, GPA 110/110 (A+)
- Main Courses: Algorithms and Data Structures, Programming, Calculus, Linear Algebra, Probability and Statistics, Logic, Databases
- Skills and technologies: C++, Java, Android, Node.js, Git, HTML/CSS, XML

Publications

- Explorer-Zero: Self-play from Environment Exploration for Long-Horizon Tasks*, **Under Submission**.
T Liu, [D Belli](#), B Major, A. Behboodi, T. Orekondy.
- When Cloud Agents Meet Device Agents: Lessons from Hybrid Multi-Agent Systems*, **ICML 2026 AIWild workshop**.
C Rainone*, [D Belli](#)*, B Major, A. Behboodi.
- Dynamic Tool Dependency Retrieval for Efficient Function Calling*, **ACL 2026** and **AAAI 2026 IR Frontiers workshop (Oral)**.
B Patel*, [D Belli](#)*, A Jalalirad, M Arnold, A Ermolov, B Major.
- Neural Augmented Kalman Filters for Road Network assisted GNSS positioning*, **ICML 2025 ML4Wireless workshop**.
H Gorp*, [D Belli](#)*, A Jalalirad, B Major.
- Efficient LLM Inference using Dynamic Input Pruning and Cache-Aware Masking*, **MLSys 2025**.
M Federici*, [D Belli](#)*, M van Baalen, A Jalalirad, A Skliar, B Major, M Nagel P Whatmough.
- GNSS Positioning using Cost Function Regulated Multilateration and Graph Neural Networks*, **ION GNSS+ 2023**.
A Jalalirad, [D Belli](#), B Major, S Jee, H Shah, W Morrison.
- Online Adaptive Personalization for Face Anti-spoofing*, **ICIP 2022**.
[D Belli](#), D Das, B Major and F Porikli.
- A Personalized Benchmark for Face Anti-spoofing*, **WACV 2022 MAP-A workshop**.
[D Belli](#), D Das, B Major and F Porikli.
- Image-Conditioned Graph Generation for Road Network Extraction*, **NeurIPS 2019 GRL workshop**.
[D Belli](#) and T Kipf.
- Adding Object Detection Skills to Visual Dialogue Agents*, **ECCV 2018 SiVL workshop**.
G Bani*, [D Belli](#)*, G Dagan*, A Geenen*, A Skliar, A Venkatesh, T Baumgartner, E Bruni and R Fernández.
- Context Encoding Chest X-rays*, **arXiv preprint**.
[D Belli](#), S Hu, E Sogancioglu and B van Ginneken.
- Chest X-ray Inpainting with Deep Generative Models*, **arXiv preprint**.
E Sogancioglu*, S Hu*, [D Belli](#) and B van Ginneken.

* = Equal contributor

Honors & Awards

2025-26	Top 12%: Codeforces contest ranking (Global)	Amsterdam, NL
2022-26	Top 4%: LeetCode contest ranking (Global)	Amsterdam, NL
2019	1st Place: Amsterdam Programming Contest	Amsterdam, NL
2018	Top 50%: ACM-ICPC Nwerc (European Semi-Finals)	Eindhoven, NL
2018	Top 50%: BAPC Regionals	Louvain-la-Neuve, BE
2016	Bronze Medal: ACM-ICPC Swerc (European Semi-Finals)	Porto, PT
2016-19	Top 15%: Google Hashcode	Amsterdam, NL
2008-13	Top 0.1% (7th out of 9000): International Championships of Mathematical Games (Italian Finals)	Milan, IT

Volunteering

- 2020-26 **Conference Reviewer:** WACV (2022-2023-2024-2025), LoG (2023-2024-2025), ICCV (2021)
- 2020-26 **Workshop Reviewer:** ML on Graphs (GLF @ NeurIPS 2022-2023, GLB @ WWW 2022-2023, GRL+ @ ICML 2020)
- 2019-26 **Mentor at LeadTheFuture:** mentorship organization for selected top students in STEM from Italy.
- 2017-19 **Member and Treasurer at Master Committee:** organizing educational events for master students at UvA.
- 2015-17 **School Tutor:** teaching Maths, Physics and Computer Science to High School and Bachelor students.

Patents

Filed 13 patent applications (3 granted, 10 pending):

- Self-play for Long-Horizon Tasks with Subtask Supervision. *US Patent App. 64/142,292, 2026.*
- Optimizing weighted least square (wls) inputs to improve global navigation satellite systems (gnss) localization. *US Patent 12,607,753, 2026.*
- Efficient Hybrid Multi-Agent Systems. *US Patent App. 64/055,449, 2026.*
- Optimizing weighted least square (wls) inputs to improve global navigation satellite systems (gnss) localization. *US Patent 12,607,753, 2026.*
- Personalized biometric anti-spoofing protection using machine learning and enrollment data. *European Patent EP4320606, 2026.*
- Architectures and Representations for Agentic AI. *US Patent App. 19/440,522, 2026.*
- System and Method of Tool Dependency Learning for Function Calling AI Agents. *US Patent App. 19/458,639, 2026.*
- System and Method for Efficient Plan Sampling From Function Calling LLMs. *US Patent App. 19/406,680, 2025.*
- Deep Learning methods for Road Network assisted GNSS positioning. *US Patent App. 19/299,160, 2025.*
- Test-time Scaling for Single-turn Function Calling. *US Patent App. 19/270,192, 2025.*
- Adaptive personalization for anti-spoofing protection in biometric authentication systems. *US Patent 12,314,365, 2025.*
- Geometric representation and temporal modeling for GNSS localization. *US Patent App. 19/040,909, 2025.*
- Cache-aware dynamic module selection. *US Patent App. 18/902,554, 2025.*
- Global navigation satellite systems (gnss) localization with residual grid representation. *US Patent App. 18/630,717, 2025.*